Samba Server

Task 1: Install Samba and create shared folder

Run as root on Linux:

dnf install samba samba-client -y

mkdir -p /srv/music-share

chmod 2770 /srv/music-share

Task 2: Create users and set Samba passwords

useradd partyuser

useradd guestuser

passwd partyuser

passwd guestuser

smbpasswd -a partyuser

smbpasswd -a guestuser

Enter passwords when prompted. The passwd commands set Linux passwords; smbpasswd sets Samba passwords.

Task 3: Configure smb.conf

Open the configuration file:

nano /etc/samba/smb.conf

Add this section at the end:

```
[music-share]
   path = /srv/music-share
   browseable = yes
   valid users = partyuser guestuser
   read only = yes
   write list = partyuser
   create mask = 0660
   directory mask = 0770
```

Save and exit (Ctrl+O, Enter, Ctrl+X in nano).

Set ownership and SELinux permissions:

chown -R root:partyuser /srv/music-share

usermod -aG partyuser guestuser

setsebool -P samba_export_all_rw on

The write list setting gives partyuser write access while guestuser remains read-only. The group commands above are not required for Samba authentication, and you should avoid relying on group membership alone to enforce the per-user access difference.

Check the configuration and start Samba:

testparm

systemctl enable --now smb

systemctl restart smb

If Windows cannot connect, allow Samba through the firewall:

firewall-cmd --permanent --add-service=samba

firewall-cmd --reload

Task 4: Connect from Windows and copy an MP3

On Linux, find the server IP:

hostname -I

On Windows:

1. Press Win + R.
2. Enter \\SERVER_IP\music-share, replacing SERVER_IP with your Linux IP.
3. Log in as partyuser with the Samba password.
4. Copy a sample .mp3 file into the folder.

Verify on Linux:

ls -l /srv/music-share

Your MP3 file should appear in the output.

Task 5: Restart Samba and check connected clients

On Linux:

systemctl restart smb

smbstatus

Look for the connected client and share name music-share.

Important: Run smbstatus while the Windows machine is connected to the share. If no client appears, open the share from Windows again and rerun the command.